



**Department of Computer Science
Faculty of Informatics
Addis Ababa University**

Entrance Examination for the M. Sc. Program in Computer Science

Date: July 13, 2007
Time allowed: 3 hours

Name: _____

Instructions

1. Read the instructions and the questions carefully.
2. Write your name **only** on this page. Writing your name on the other pages will automatically disqualify you from this exam.
3. Answer all questions eligibly on the spaces provided on the question papers. Please make sure that sentences are short, and grammatically and logically correct. You can use the back side of each paper when space is not sufficient. In such a case, please indicate so.
4. This examination booklet consists of **11** pages (excluding this cover page). Make sure that the question papers are numbered consecutively (at the bottom of each page) and that no question papers are missing or replicated.
5. Cheating is strictly prohibited. Any person found cheating will be automatically disqualified from the exam.
6. Use only the rough papers provided. Using other unauthorized papers will be considered as cheating.
7. Switch **off** your mobile phones. You are not allowed to use a mobile for **any** purpose.
8. You are not allowed to use calculators or any other electronic device.

For Office Use Only

Part	Part I (35)	Part II (25)	Part III (20)	Part IV (20)	Total (100)
Marks					

Part I: Programming, Data Structures, and Software Engineering (35 points)

- 1.1

a.

What is a software development process model?

(2 points)
- b.

Explain in detail the Rapid Application development model.

(3 points)
- 1.2

a.

What is software risk?

(2 points)

b. Discuss software process risk. (3 points)

1.3 a. What is the difference between in-fix notation and post-fix notation? (1 point)

b. Convert the expression $A + (B * C - D/E * F) * H$ into post-fix notation. (3 points)

1.4 Given a binary tree BT, write an algorithm (using pointer implementation) that counts the number of nodes. (5 points)

- 1.5 Rewrite the following recursive algorithm using iterative methods using a programming language of your choice. You may introduce new variables as required. (5 points)

```
int f(int num)
Begin
/* num > 0 */
  if( num <= 2 ) then
    return 1;
  else
    return f(num - 1) + f(num - 2);
  end if
End
```

- 1.6 Mention at least three points in which object oriented programming is advantageous than structured programming. (3 points)

- 1.7 Write three benefits of modular programming? (3 points)

- 1.8 Write the implementation of a simple sorting algorithm with a programming language of your choice to order a list having n integer values in descending order. (5 points)

Part II: Computer Architecture and Operating Systems (25 points)

2.1 In the 16-bit FLAGS register, 9 of the 16-bits are used as flags. What are the two groups in which these flag bits are grouped? Describe the purpose of each group. (4 points)

2.2 Compare CISC and RISC in light of emphasis on the following points? (5 points)

Point of Emphasis	CISC	RISC
Program code size and clock cycle needed		
“LOAD” and “STORE” instructions		
Transistors much needed for		
Hardware or Software emphasis		
Instructions’ complexity		

2.3 In von Neumann architecture, the scheme of segment plus offset allows a program code or data to be relocate able. Describe in your words what relocation is and how it is done. (4 points)

- 2.4 In most modern operating systems, making use of threads in a single process is common. Write two basic reasons (advantages) for using threads. (2 points)
- 2.5 Direct Memory Access (DMA) is commonly available in general-purpose modern computer systems. What are the disadvantages of a system without a DMA? (2 points)
- 2.6 What are the four types of files supported by the Unix operating system? (2 points)
- 2.7 Paging and segmentation are two common methods of implementing virtual memory. Unlike pages, segments do not have fixed size and have independent linear address space. Give two advantages of segmentation over paging. (3 points)
- 2.8 What is the Master Boot Record (MBR) in a file System? What happens to the system if the MBR is destroyed, say by a virus? (3 points)

3.1 Consider the following six relations for an order processing database application in a company:

WAREHOUSE(WarehouseNo, City)

Write SQL statements for the following queries:

- Page -7-*

c. Produce a listing: **CUSTNAME**, **NO_OF_ORDERS**, **AVG_ORDER_AMT**, where the **NO_OF_ORDERS** is the total number of orders by the customer and the **AVG_ORDER_AMT** is the average order amount for that customer. (3 points)

d. List the orders that were not shipped within 30 days of ordering. (1.5 points)

e. List the **OrderNo** for orders that were shipped from **all** warehouses that the company has in '**Addis Ababa**'. (3 points)

3.2 What is the difference between logical data independence and physical data independence? (3 points)

3.3 How do we transform n-ary relationships between entities in E-R diagram into relations? Give an example. (3 points)

3.4 Is a relation in 3NF also in BCNF? Justify your answer and give a relevant example. (4 points)

Part IV: Computer Networks and Data Communications (20 points)

4.1 For each of the following questions, circle the letter of the correct answer. (1 point each)

- i. A set of rules governing communication within and between computing end points is described as
 - a. Point-to-point b. Protocol c. Medium d. Duplex
- ii. To which layer(s) of the IEEE 802 does the data link layer of the OSI/ISO reference model map?
 - a. Physical b. LLC c. MAC d. a and b e. b and c
- iii. The following are functions of a router except
 - a. Traffic isolation b. Broadcast storm protection
 - c. Optimal routing d. Internetworking e. None of the above
- iv.. Networks in a company are best described as
 - a. Internet b. internet c. Intranet d. Extranet
- v. Which of the following OSI/ISO layer relieves other upper layers from the burden of knowing the attached internetworking devices such as Routers, Switches, etc.?
 - a. Application layer b. Data Link layer
 - c. Network Layer d. Transport layer e. Physical layer
- vi. In a TCP/IP peer-to-peer network, which of the following layer(s) is not required?
 - a. Data link layer b. Internet/Internetworking layer
 - c. Physical layer d. None of the above e. All of the above
- vii. An application layer address is called
 - a. MAC address b. Port number c. Host name
 - d. IP address e. All of the above
- viii. Which of the following layer(s) is involved in the sending and receiving hosts as well as in the intermediary nodes such as routers?
 - a. Application layer b. Network layer
 - c. Transport layer d. None of the above e. All of the above

4.2. Mention one mechanism which is used in a data communication network to detect error during transmission _____ (1 point)

4.3 Mention and briefly describe two media access protocols. (2 points)

4.4 What is the addressing mechanism used in Ethernet LAN? Briefly describe. (2 points)

4.5 Mention and briefly describe at least two major reasons why IPv6 was defined to replace IPv4. (2 points)

4.6 In comparing circuit switching and packet switching, write **Yes** or **No** in the last two columns. (3 points)

Item	Circuit Switching	Packet Switching
Dedicated physical path required		
Each packet follows the same route		
Packets arrive in order		
A switch crash is fatal		
Potentially wasted bandwidth		
Store-and-forward transmission		

4.7 Mention and briefly describe two attenuation impairments in data transmission. (2 points)